

MIT OPENCOURSEWARE ÇÖZÜMLÜ SORULARI -2

<https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/>

<http://web.mit.edu/6.003/F11/www/handouts/hw8-solutions.pdf>

Soru – 1: Verilen $x(t)$ işaretlerinin Fourier seri katsayılarını bulunuz.

P7.3

Find the Fourier series coefficients for each of the following signals:

(a) $x(t) = \sin\left(10\pi t + \frac{\pi}{6}\right)$

(b) $x(t) = 1 + \cos(2\pi t)$

(c) $x(t) = [1 + \cos(2\pi t)] \left[\sin\left(10\pi t + \frac{\pi}{6}\right) \right]$

Hint: You may want to first multiply the terms and then use appropriate trigonometric identities.

Çözüm -1:

S7.3

(a)
$$x(t) = \sin\left(10\pi t + \frac{\pi}{6}\right)$$
$$= \frac{e^{j\pi/6}}{2j} e^{j2\pi t5} - \frac{e^{-j\pi/6}}{2j} e^{-j2\pi t5}$$

We choose ω_0 , the fundamental frequency, to be 2π .

$$x(t) = \sum_k a_k e^{jk\omega_0 t},$$

where

$$a_5 = \frac{e^{j\pi/6}}{2j}, \quad a_{-5} = \frac{-e^{-j\pi/6}}{2j}$$

Otherwise $a_k = 0$.

(b)
$$x(t) = 1 + \cos(2\pi t)$$
$$= 1 + \frac{e^{j2\pi t}}{2} + \frac{e^{-j2\pi t}}{2}$$

For $\omega_0 = 2\pi$, $a_{-1} = a_1 = \frac{1}{2}$, and $a_0 = 1$. All other a_k 's = 0.

(c)
$$x(t) = [1 + \cos(2\pi t)] \left[\sin\left(10\pi t + \frac{\pi}{6}\right) \right]$$
$$= \sin\left(10\pi t + \frac{\pi}{6}\right) + \cos(2\pi t) \sin\left(10\pi t + \frac{\pi}{6}\right)$$
$$= \left(\frac{e^{j\pi/6}}{2j} e^{j2\pi t5} - \frac{e^{-j\pi/6}}{2j} e^{-j2\pi t5} \right) + \left(\frac{1}{2} e^{j2\pi t} + \frac{1}{2} e^{-j2\pi t} \right) \left(\frac{e^{j\pi/6}}{2j} e^{j2\pi t5} - \frac{e^{-j\pi/6}}{2j} e^{-j2\pi t5} \right)$$
$$= \frac{e^{j\pi/6}}{2j} e^{j2\pi t5} - \frac{e^{-j\pi/6}}{2j} e^{-j2\pi t5} + \frac{e^{j\pi/6}}{4j} e^{j2\pi t6} - \frac{e^{-j\pi/6}}{4j} e^{-j2\pi t4}$$
$$+ \frac{e^{j\pi/6}}{4j} e^{j2\pi t4} - \frac{e^{-j\pi/6}}{4j} e^{-j2\pi t6}$$

Therefore,

$$x(t) = \sum_k a_k e^{jk\omega_0 t},$$

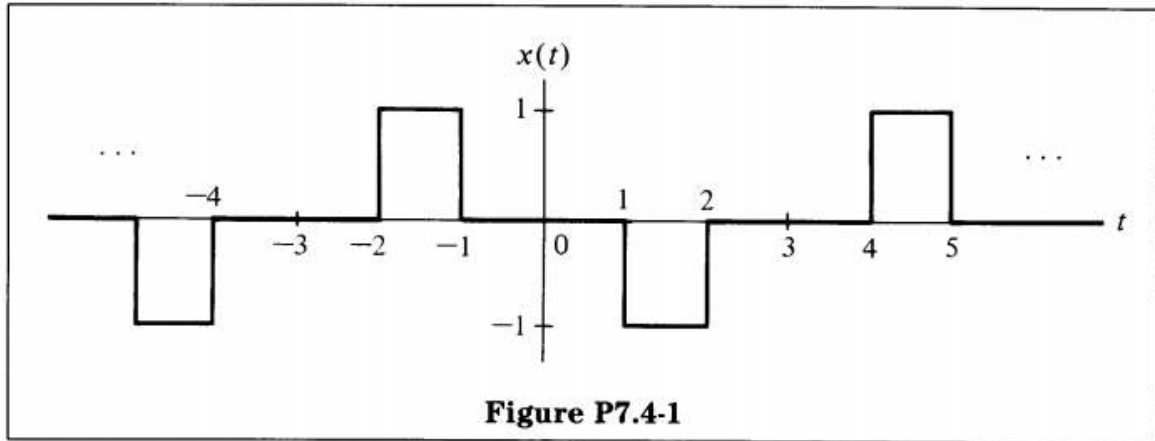
where $\omega_0 = 2\pi$.

$$\begin{aligned} a_4 &= \frac{e^{j\pi/6}}{4j}, & a_{-4} &= \frac{-e^{-j\pi/6}}{4j}, \\ a_5 &= \frac{e^{j\pi/6}}{2j}, & a_{-5} &= \frac{-e^{-j\pi/6}}{2j}, \\ a_6 &= \frac{e^{j\pi/6}}{4j}, & a_{-6} &= \frac{-e^{-j\pi/6}}{4j} \end{aligned}$$

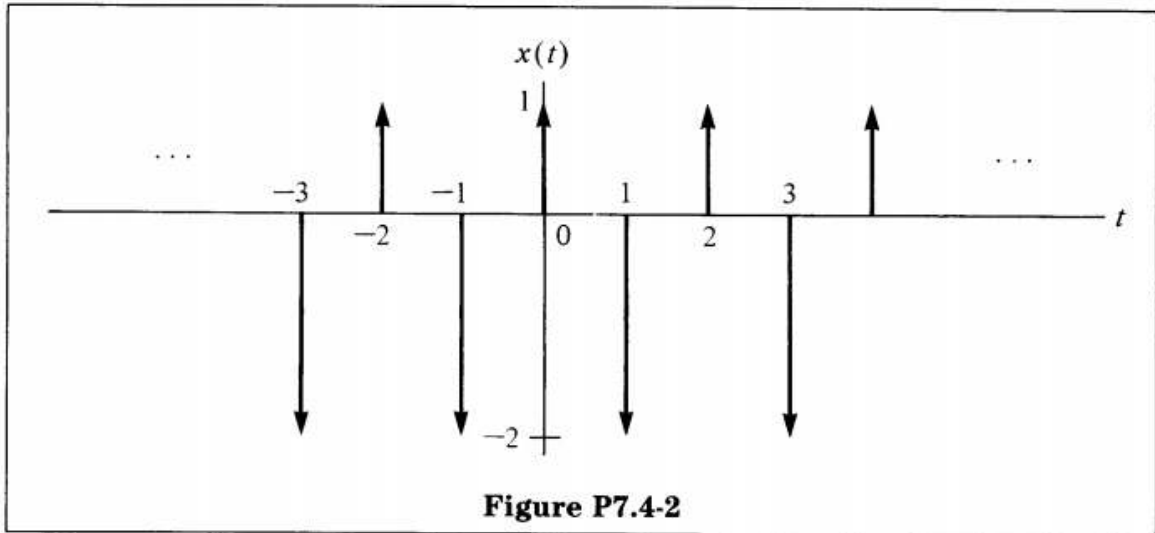
All other a_k 's = 0.

Soru – 2: Fourier seri analiz denklemini kullanarak aşağıdaki işaretlerin seri katsayılarını bulunuz.

(a)

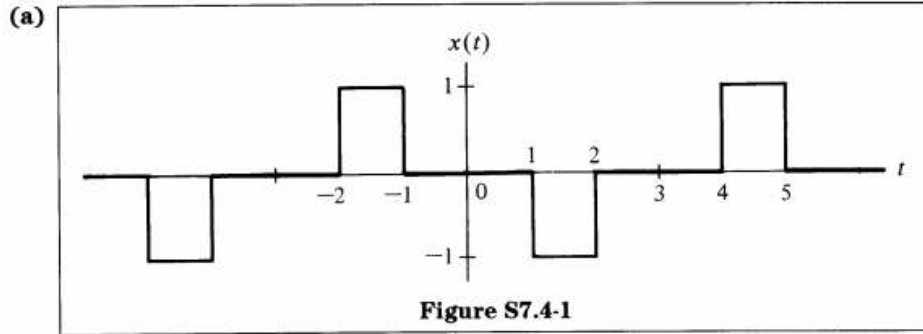


(b)



Çözüm -2:

S7.4



Note that the period is $T_0 = 6$. Fourier coefficients are given by

$$a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

We take $\omega_0 = 2\pi/T_0 = \pi/3$. Choosing the period of integration as -3 to 3 , we have

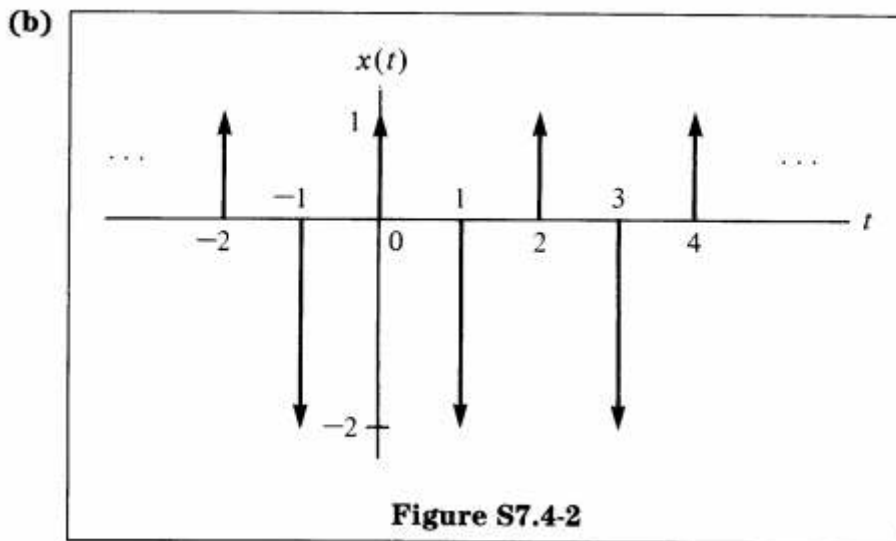
$$\begin{aligned} a_k &= \frac{1}{6} \int_{-2}^{-1} e^{-jk(\pi/3)t} dt - \frac{1}{6} \int_1^2 e^{-jk(\pi/3)t} dt \\ &= \frac{1}{6} \frac{1}{-jk(\pi/3)} e^{-jk(\pi/3)t} \Big|_{-2}^{-1} - \frac{1}{6} \frac{1}{-jk(\pi/3)} e^{-jk(\pi/3)t} \Big|_1^2 \\ &= \frac{1}{-j2\pi k} [e^{+j(\pi/3)k} - e^{+j(2\pi/3)k} - e^{-j(2\pi/3)k} + e^{-j(\pi/3)k}] \\ &= \frac{\cos(2\pi/3)k}{j\pi k} - \frac{\cos(\pi/3)k}{j\pi k} \end{aligned}$$

Therefore,

$$x(t) = \sum_k a_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{\pi}{3}$$

and

$$a_k = \frac{\cos(2\pi/3)k - \cos(\pi/3)k}{j\pi k}$$



The period is $T_0 = 2$, with $\omega_0 = 2\pi/2 = \pi$. The Fourier coefficients are

$$a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

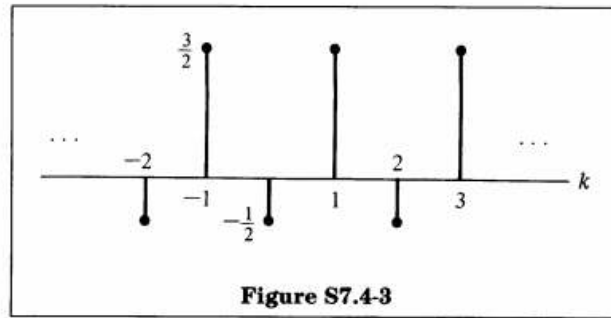
Choosing the period of integration as $-\frac{1}{2}$ to $\frac{3}{2}$, we have

$$\begin{aligned} a_k &= \frac{1}{2} \int_{-1/2}^{3/2} x(t) e^{-jk\omega_0 t} dt \\ &= \frac{1}{2} \int_{-1/2}^{3/2} [\delta(t) - 2\delta(t-1)] e^{-jk\omega_0 t} dt \\ &= \frac{1}{2} - e^{-jk\omega_0} = \frac{1}{2} - (e^{-j\pi})^k \end{aligned}$$

Therefore,

$$a_0 = -\frac{1}{2}, \quad a_k = \frac{1}{2} - (-1)^k$$

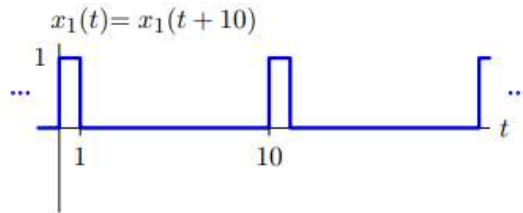
It is instructive to plot a_k , which we have done in Figure S7.4-3.



Soru – 3: Aşağıdaki $x_1(t)$ sinyalinin Fourier Seri katsayılarını bulunuz.

1. Fourier Series

Determine the Fourier series coefficients a_k for $x_1(t)$ shown below.



Çözüm – 3:

$$a_k = \frac{1}{T} \int_T x(t) e^{-j\frac{2\pi}{T}kt} dt = \frac{1}{10} \int_0^1 1 e^{-j\frac{2\pi}{10}kt} dt = \frac{1}{10} \left. \frac{e^{-j\frac{\pi}{5}kt}}{-j\frac{\pi}{5}k} \right|_0^1 = \frac{1}{j2\pi k} (1 - e^{-j\pi k/5})$$

Notice that this expression is badly formed at $k = 0$. We could use l'Hôpital's rule to evaluate this expression, but an easier method (which is also more robust against errors) is to simply evaluate the average value of $x_1(t)$ to find that $a_0 = 1/10$.

This solution could also be written in terms of sinusoids as

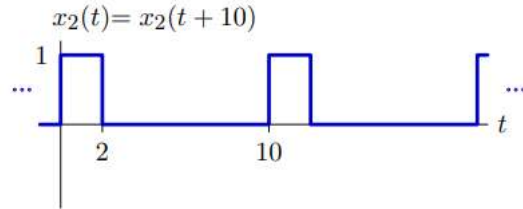
$$a_k = \begin{cases} \frac{1}{10} & k = 0 \\ \frac{1}{\pi k} e^{-j\pi k/10} \sin(\pi k/10) & k \neq 0 \end{cases}$$

$$a_0 = \frac{1}{10}$$

$$a_k = \frac{1}{\pi k} e^{-j\pi k/10} \sin(\pi k/10) \quad \text{for } k \neq 0$$

Soru – 4: Aşağıdaki $x_2(t)$ sinyalinin Fourier Seri katsayılarını bulunuz.

Determine the Fourier series coefficients b_k for $x_2(t)$ shown below.



Çözüm – 4:

$$b_k = \frac{1}{T} \int_T x(t) e^{-j\frac{2\pi}{T}kt} dt = \frac{1}{10} \int_0^2 1 e^{-j\frac{2\pi}{10}kt} dt = \frac{1}{10} \left. \frac{e^{-j\frac{\pi}{5}kt}}{-j\frac{\pi}{5}k} \right|_0^2 = \frac{1}{j2\pi k} (1 - e^{-j2\pi k/5})$$

As with the previous part, this expression is badly formed for $k = 0$. We therefore obtain $b_0 = 1/5$ by calculating the average value of $x_2(t)$.

This solution could also be written in terms of sinusoids as

$$b_k = \begin{cases} \frac{1}{5} & k = 0 \\ \frac{1}{\pi k} e^{-j\pi k/5} \sin(\pi k/5) & k \neq 0 \end{cases}.$$

$$b_0 = \frac{1}{5}$$

$$b_k = \frac{1}{\pi k} e^{-j\pi k/5} \sin(\pi k/5) \quad \text{for } k \neq 0$$